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(57) Abstract :
ABSTRACT SMART AGRICULTURE AND WEATHER PREDICTION SYSTEM USING IOT FOR DROUGHT-PRONE REGIONS The present invention discloses a Smart Agriculture and Weather Prediction System (100) using Internet of Things technology for real-time monitoring of agricultural field conditions in drought-prone regions. The system comprises an ESP32 microcontroller module (110) connected to a DHT11 temperature and humidity sensor (120), a soil moisture sensor (130), and a smoke detection sensor (140). The sensors collect environmental data including ambient temperature, relative humidity, soil water content, and smoke or harmful gas levels. The ESP32 microcontroller (110) processes the sensor data and transmits it wirelessly via integrated Wi-Fi communication module (150) to a cloud server platform (160) utilizing Blynk IoT services. A dashboard and user interface module (170) provides visual representation of collected data accessible through web and mobile applications. An alert generation module (180) automatically generates notifications when sensor readings exceed predefined thresholds for irrigation scheduling, fire safety, and crop health monitoring. The invention provides a cost-effective, energy-efficient, and farmer-friendly solution for precision agriculture enabling data-driven decision-making in drought-prone agricultural regions.

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